

NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Sixth Semester B.E. (E&E) (Credit System) Degree Examinations

May - June 2023

20EEE161 - HYBRID ELECTRIC VEHICLES

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

Unit – I		Marks	BT*	CO*	PO*
1.	a) List the functions of power train in a vehicle and explain the concept of hybrid electric drive trains.	8	L*2	2	1,2,3,4
	b) Discuss possible EV configurations due to the variations in electric propulsion characteristics and energy sources.	6	L2	1	1,2,3,4
	c) Mention the advantages and disadvantages of series hybrid electric configuration.	6	L2	1	1,3,6
2.	a) Briefly explain the factors to be considered involved in power rating design of PPS in series hybrid electric vehicles.	6	L3	2	1,2,3,4
	b) Discuss the salient features of commonly used mechanical torque coupling devices in parallel hybrid electric drive trains.	6	L2	1	1,3,6
	c) Compare the configurations of the DC/DC converter on the PPS side with that on the engine/generator side.	8	L3	2	1,2,3,4
3.	a) Explain the various potential operation modes of Series hybrid electric drive trains.	6	L2	2	1,3,6
	b) Draw a neat Control flowchart of the Max. SOC-of-PPS control strategy and explain.	8	L4	2	1,2,3,4
	c) Explain in detail tractive effort and transmission requirement in consideration with single-gear and multi-gear transmission.	6	L4	1	1,3,6
Unit – II					
4.	a) Explain the terms: Charge-Depleting Range (CDR) and PHExx.	4	L2	4	1,2,6
	b) With a neat diagram explain the overall control scheme of the parallel hybrid drive train.	6	L2	3	1,2,6
	c) Analyse the various modes of operation in series – parallel drive train in speed- coupling and torque coupling mode.	10	L2	3	1,2,6
5.	a) Discuss the two energy management strategies used in Plug-In Hybrid Electric Vehicles.	8	L2	3	1,2,6
	b) Explain the Engine Turn-On and Turn-Off (Engine-On-Off) Control Strategy used in parallel hybrid electric drive train.	6	L2	3	1,2,6
	c) Explain the operation of planetary gear unit used as a speed-coupling device.	6	L2	3	1,2,6
6.	a) Briefly explain the energy storage design (PPS) in parallel hybrid electric drive train.	8	L4	3	1,2,6
	b) Discuss the most important parameters in the energy storage design of Plug-In Hybrid Electric Vehicles.	6	L2	4	1,2,6
	c) Discuss the Engine Speed Control Strategy in low, medium and high-speed operation of a Series- Parallel Hybrid Drive Train.	6	L1	3	1,2,6

P.T.O.

Unit – III

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| 7. a) | Discuss the various aspects of design and control principles for Maximum Regenerative Braking. | 6 | L1 | 5 | 1,3 |
| b) | With a neat schematic structure explain the parallel-hybrid brake system. | 6 | L2 | 5 | 1,3 |
| c) | Explain four classifications of CAN message frame structures in a CAN protocol. | 8 | L2 | 5 | 1,3 |
| 8. a) | Explain the control strategy for optimal braking performance. | 6 | L2 | 5 | 1,3 |
| b) | What is Bus utilization in CAN and how to calculate bus utilization? | 6 | L2 | 5 | 1,3 |
| c) | Analyze how braking energy and vehicle speed are related for the design and control of the regenerative brake system. | 8 | L4 | 5 | 1,3 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
